IVF Digital Twin

Version 7.1 · Technical Documentation

A Dynamic AI Framework for Reproductive Potential Assessment and Continuous IVF Outcome Monitoring

Stage simulation · Neural ensemble · Diffusion cross-check · Patient-similarity graph · Bayesian evidence fusion · Early warning · MCP agents

From in vitro to in silico

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Future Health Digital Innovation Challenge 2026 · Fertility · Reproductive Potential

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Research and clinical decision-support software. Not a medical device. Outputs must be reviewed by a qualified clinician.

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Document control

Item	Value
Document	IVF Digital Twin: Technical Documentation, version 7.1
Supersedes	IVF Digital Twin v7.0 technical documentation (July 2026)
Software release	IVF Digital Twin 7.1, application revision 2026-09-13.1; model fingerprint 91704e29 (locked release used for validation)
Companion components	DT × Laya Android app 1.3.0; MCP connector `ivf_digital_twin`; Early Warning System for embryo culture; laboratory analytics pipeline `embryo-lab-data`
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Contact	embryossa@gmail.com · https://github.com/embryossa/IVF-Digital-Twin
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Regulatory status	Research and clinical decision-support software. Not a medical device; not registered, cleared or approved as one in any jurisdiction.

> How to read this document. Sections 1 to 3 describe purpose, users and architecture. Sections 4 to 7 specify the models and modules as implemented in release 7.1. Section 8 describes how the system addresses the Reproductive Potential challenge. Sections 9 and 10 report validation and software verification. Sections 11 to 13 cover security, responsible AI, and the assumptions and limitations required by the challenge rules. Section 14 lists the changes from version 7.0.

1. Summary

The IVF Digital Twin (DT) is a continuously updated computational representation of one person's reproductive potential and of an individual IVF treatment cycle. It does not return a single "fertility score". It returns distributions: the expected number of retrieved oocytes, mature (MII) oocytes, two-pronuclear (2PN) zygotes, blastocysts, good-quality blastocysts and euploid embryos, and the probability of clinical pregnancy per transfer, per cycle and across a multi-cycle plan. Each estimate carries a model-uncertainty range, a reliability index, out-of-distribution flags and an explicit record of which inputs were measured, missing, observed as zero or imputed.

The same engine runs at four moments of one person's pathway:

- **Before treatment:** age, AMH, AFC, BMI and history produce expected yields. Illustrative scenario projections (treatment now, delayed treatment, several cycles, oocyte cryopreservation) come from the Total Reproductive Potential (TRP) and banking modules. These are model-based scenarios, not validated individual forecasts of future fertility or live birth.
- **After oocyte retrieval:** observed counts replace their simulated values. The earlier forecast is retained and downstream expectations are updated by conditional simulation.
- **During culture:** observed 2PN, day-3 and blastocyst counts are compared with frozen forecast envelopes. Material deviations are flagged for the embryologist.
- **At laboratory and system level:** the expected yields for the case mix actually treated become a reference for laboratory key performance indicators (KPIs). This reference feeds statistical early warning and AI-agent explanations through the Model Context Protocol (MCP).

Release 7.1 is a working product. It comprises a Windows desktop application, a Windows service with a REST API for clinic information systems, an MCP connector for AI agents, an offline Android app and a laboratory early-warning system. All computation runs on clinic hardware or on the phone, without internet access.

Key validated results of the locked release (section 9):

Evaluation	Result
Prospective clinical cohort (centre D, forecasts disclosed only after transfer)	AUC 0.761 (95% CI 0.629–0.878), Brier 0.213, 59 transfers
Temporal validation at a new centre (centre E, 2026)	AUC 0.702 (0.622–0.771), 234 fresh transfers, locked model without local training; models trained on 1,610 of the centre's own transfers reached 0.656–0.743 (logistic regression 0.732)
Stage forecasting from oocyte and MII counts	Median absolute error one embryo per stage; 2PN ρ = 0.961; blastocysts ρ = 0.704; good blastocysts ρ = 0.732; 88–98% of counts inside the 95% simulation interval
Calibration	Prospective cohort over-predicted by 8.9 percentage points without local recalibration; a centre intercept restored O/E to 0.99–1.02 at centre E
Added value of 114 extended clinical variables	None for the DT (likelihood-ratio p = 0.80; no block improved the 2026 test)
Added value of embryo images	No consistent gain; explanation maps unstable

2. Intended use, users and scope

2.1 Intended use

The DT provides probabilistic, explainable estimates of reproductive output and clinical-pregnancy probability for use by qualified reproductive-medicine professionals in counselling, treatment planning and laboratory quality management. Clinical pregnancy is defined as fetal cardiac activity on ultrasound. Live birth is not modelled.

2.2 Users

User	Use	Interface
Reproductive-medicine clinician	Counselling before and during treatment; banking and multi-cycle planning	Desktop app, PDF report (EN/RU), Android app
Embryologist and laboratory director	Expected-versus-observed review per cycle and per period; early warning; incubator and media-lot review	Desktop app, early-warning system, laboratory analytics report
Clinic information system	Automated submission of cycles and retrieval of results after clinician review	REST API v1
AI agent under clinic control	Batch preparation, scoring and explanation of results with pseudonymised data	MCP connector
Health-system analyst (pilot)	Aggregated, case-mix-adjusted monitoring	Aggregated exports and dashboard (planned)
Person planning a family or patient	Receives explanations through the clinician; no direct self-service	Clinician-mediated report

2.3 Out of scope and prohibited uses

- Autonomous diagnosis, treatment prescription or medication dosing. The stimulation section is a reference summary, not a prescription.
- Embryo ranking within a cohort from images. The DT has no image input.
- Use of outputs for employment, insurance, eligibility or any decision unrelated to the person's own care. The pilot licence prohibits this.
- Use without local calibration review in a population that differs materially from the development cohorts (section 13).

3. System architecture

3.1 Overview

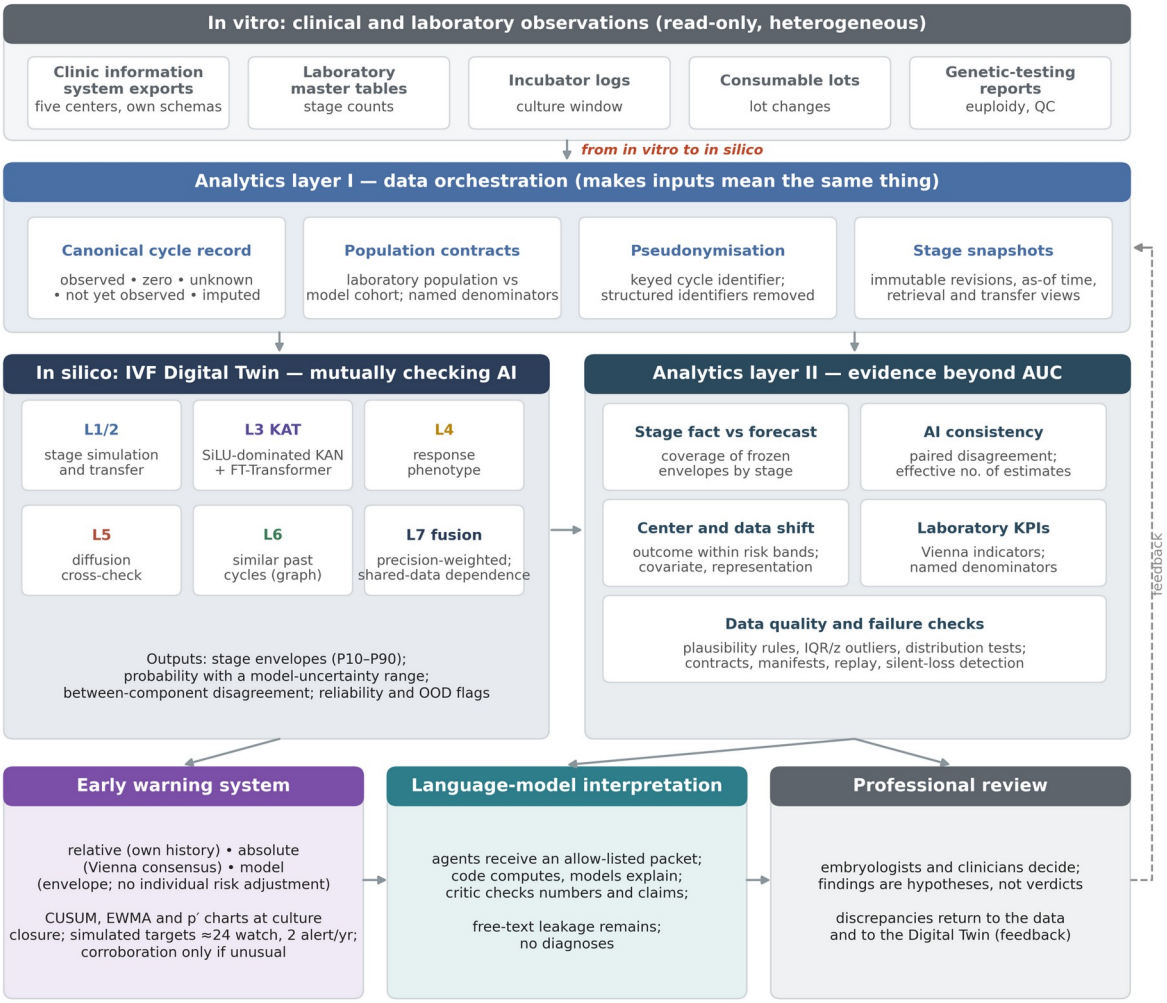


Figure 1. From laboratory observations to reviewable Digital Twin evidence. Read-only records become canonical cycle states with an explicit observation status, population definitions, pseudonyms and stage snapshots. The DT returns stage envelopes, pregnancy estimates and reliability indicators. The analytics layer examines these outputs and the data that produced them. Laboratory monitoring keeps local, consensus and model references separate. Language models explain allow-listed computed packets under professional review.

The system has three tiers:

- **Analytics layer I, data orchestration.** Converts heterogeneous clinic exports into canonical cycle states. Every value carries one of five observation states: measured, observed zero, unknown, not yet observed at the prediction time, or imputed (flagged). The layer defines laboratory and model populations with named denominators, assigns keyed-hash pseudonyms, and stores immutable, time-stamped stage snapshots.
- **In silico core, the seven-layer Digital Twin (section 4).** It computes stage envelopes, pregnancy probabilities, an uncertainty range, a reliability index and out-of-distribution (OOD) flags.
- **Analytics layer II, evidence beyond AUC.** Evaluates outputs and data before they reach laboratory monitoring or language-model interpretation. It covers stage fact versus forecast, agreement between models, centre and data shift, Vienna-consensus KPIs and data-quality checks.

3.2 Deployment components (release 7.1)

Component	Description	Runtime
Desktop application	Five sections (New calculation, Results, History, Incoming cycles, Setup); clinical, extended and research views; PDF report in English and Russian; CSV/XLSX batch input	Windows 10/11 x64, bundled Python 3.11.5 runtime, Edge WebView2; CPU only
Single computation path	`desktop/service.py::Service.compute` serves manual entry, batch runs, the REST API and MCP. No alternative algorithm or calibration exists for any interface	Same
Clinic integration service	Windows service (LocalService), HTTPS REST API v1, encrypted inbox of incoming cycles, clinician-review workflow, analytics scope for research batches	Loopback or private IPv4 only
MCP connector `ivf_digital_twin`	11 tools for AI agents (Claude Code, Codex, Claude Desktop): table inspection, cycle preparation, submission, results, reports, export. Standard library only	Bundled Python
Local language model	Gemma 4 via the bundled Ollama engine, for narration of computed results only	Local; downloaded once at setup
DT × Laya Android app 1.3.0	On-device DT layers 1–3 (2,000 reproducible scenarios), Laya encoder for text intake, TRP outlook, encrypted history	Android 10+, ARM64; no INTERNET permission
Early Warning System	Daily process control of laboratory KPIs against three references	Python 3.10+, local Teable
Laboratory analytics pipeline	Vienna-consensus KPIs, model reconciliation, culture-window forensics, pseudonymised agent review, HTML/XLSX report	Python; MIT licence

3.3 Data flow and observation states

A cycle record is a sequence of stage snapshots: baseline, retrieval, fertilisation, day 3, blastocyst, transfer. Each snapshot is immutable once accepted. New revisions are appended and retain their history. The integration service accepts duplicates idempotently and rejects conflicting, out-of-order, future-dated or outcome-bearing submissions. Synthetic contract tests covered these cases: 21 of 21 returned the expected response.

The DT distinguishes an empty field (stage unknown) from an observed zero (stage known to have failed). Known stage counts constrain the joint simulation. Downstream stages are sampled conditionally and are not treated as additional predictions. Imputed inputs, such as a missing AMH replaced by a population value, are recorded in the run manifest, and the forecast is labelled as describing a partly hypothetical patient.

4. The seven layers of the Digital Twin (release 7.1)

4.1 Layer 1: stochastic stage simulation

Layer 1 simulates the cycle as a conditional cascade with 5,000 Monte Carlo draws per cycle in the desktop application. The Android app uses 2,000 reproducible draws with seed 42.

Stage	Model in 7.1	Source of parameters
Retrieved oocytes	Negative binomial with log mean depending on AMH, AFC and age; fitted by zero-truncated likelihood to stimulated cycles with ≥ 1 oocyte, five-fold patient-level cross-validation	Development data; coefficients updated 11 Sep 2026
MII oocytes, 2PN zygotes	Binomial transitions with per-oocyte logistic coefficients	Imported unchanged from Craig et al., J Assist Reprod Genet 2026
Blastocysts	Zero-inflated beta-binomial; mean depends on age and number of embryos entering the stage	7,919 culture-observed cycles, centres A–C
Good-quality blastocysts (≥ 3BB)	Beta-binomial	Same
Euploid embryos	Age-stratified prior with explicit dispersion	Published aneuploidy and mosaicism data (Franasiak 2014; Armstrong 2023)
Warming survival	Fixed probability 0.95	Literature
Transfer quality	Fair- vs good-quality blastocyst adjusted OR 0.51 (95% CI 0.35–0.75)	2,975 day-5 single-blastocyst transfers
Several transfers from one retrieval	Shared cycle-level logit-normal effect, σ = 0.746	Clinic exports with recorded outcome of every transfer

Without PGT-A, every blastocyst that survives warming is available for transfer, good-quality embryos first. Entering a number of euploid embryos switches on the PGT-A scenario. PGT-A is modelled as a scenario of embryo selection and use. It does not change biological reproductive potential, and the model does not imply that it increases cumulative live birth. The "transfers planned" field limits the per-cycle probability to that number of sequential transfers. It does not alter the per-transfer probability or the neural-network inputs.

Explicit risk outputs per cycle include the share of scenarios without any blastocyst, without a good-quality blastocyst and without an embryo for transfer. Oocyte-count thresholds are associated with OHSS risk.

4.2 Layer 2: per-transfer ensemble

Two per-transfer estimates are averaged with equal weight on the logit scale:

- **Clinical estimate.** A logistic function of standardised female age, log AMH and BMI, with a draw-level Gaussian effect (SD 0.07 on the logit scale). The centring follows the FORTUNE development cohort, and the coefficients were specified for this platform. For the local population, the intercept was recalibrated from 0.40 to −0.25 by maximum-likelihood logistic regression, with the original logit as an offset, in the 1,556-cycle calibration set.
- **Laboratory estimate.** The five-component KPI score (range 5–25; Sergeev & Diakova, Sci Rep 2024) mapped to a beta distribution moment-matched to a score-specific 95% interval of implantation probability.

The module reports the per-transfer probability, the probability conditional on an available embryo and the whole-cycle probability.

4.3 Layer 3: KAT neural core

KAT is an ensemble of a Kolmogorov–Arnold network (KAN) and an FT-Transformer trained on 18 tabular predictors:

- female age (donor age in donor-oocyte cycles)

- attempt number
- counts: follicles, cumulus–oocyte complexes, inseminated oocytes, 2PN zygotes, day-3 and day-5 embryos, blastocysts, good-quality blastocysts, cryopreserved and transferred embryos
- rates: fertilisation, cleavage, blastocyst, good-quality blastocyst and oocyte retrieval
- the laboratory KPI score

AMH and AFC are not KAT inputs; they enter through Layer 1.

Element	Specification
Data	16,056 cycles, 2013–2023, six countries; random 80/10/10 split (seed 42); held-out test 1,606 cycles (521 pregnancies)
Pre-processing	Infinite rates set to missing; median imputation; clipping at the 1st and 99th percentiles
KAN	18–10–1, B-spline grid 5, order 3, grid re-adapted every 20 epochs (pykan 0.2.1); 200 epochs full-batch AdamW (lr 1e-3, weight decay 1e-4), cosine annealing, gradient clipping 1.0
FT-Transformer	mambular; embedding 64, 8 layers, piecewise-linear encoding (50 bins); up to 150 epochs, lr 1e-4, batch 256, early stopping (patience 10)
Ensemble	Softmax weights optimised with frozen components: 0.27 KAN, 0.73 FT-Transformer
Calibration	Isotonic regression on validation-partition predictions; 7.1 interpolates between step centres
Held-out performance	AUC 0.820 (95% CI 0.798–0.840); Brier 0.163
7.1 correction	Piecewise-linear encoding now uses the trained bin boundaries rather than the maximum of the current batch. A profile scored alone or in a batch gives the same output (tolerance 1e-6)

4.4 Layer 4: response phenotypes

k-means clustering (ten restarts) of 18 standardised variables on the 1,556-cycle calibration set distinguishes poor, standard and high responders. It was compared with Ward clustering and DBSCAN; the three-cluster solution was retained (silhouette 0.186). For a new cycle the layer returns membership probabilities and the dominant phenotype. It is clinical context, not a separate pregnancy probability and not a formal POSEIDON assignment.

4.5 Layer 5: CSDI Hybrid v3 diffusion cross-check

A conditional score-based diffusion model with six transformer layers generates blastocyst and good-quality blastocyst counts from seven variables available after the day-1 fertilisation check. It was developed on 15,193 cycles from three centres, with 1,520 held out for testing.

- Inputs are quantile-normalised.
- Sampling is a deterministic 50-step DDIM, so repeated runs give the same distribution.
- Count intervals come from split conformal prediction.
- A LightGBM classifier with Venn–Abers calibration estimates pregnancy from the conditioning variables and the generated counts. Held-out AUC 0.661.

Layer 5 checks whether the embryology forecast is consistent. It is not an independent vote equal to KAT and GAT. Its agreement with Layer 1 modifies the precision of the biological source. An intentional abstention (no suitable 2PN) is distinguished from a computation error, and erroneous or out-of-domain outputs are excluded from fusion.

4.6 Layer 6: patient-similarity graph (GAT)

Each of the 16,056 development cycles is a node described by the 18 KAT predictors. Edges are built in four steps:

1. Retrieve 30 candidates per node by cosine similarity.
2. Select up to six neighbours by maximal marginal relevance ($\lambda = 0.5$).

3. Accept a neighbour when similarity is ≥ 0.72, or while the node has fewer than two edges.
4. Symmetrise the edge weights.

The graph transformer has three pre-normalisation residual TransformerConv blocks (four heads, 64 dimensions) and 107,890 parameters. It was trained semi-supervised with label smoothing and a class weight. For a new cycle the ten most similar training cycles are found and the neighbourhood is rebuilt with the new, unlabelled node. Graph support combines three quantities: the effective number of neighbours, the agreement of neighbour predictions and the entropy of neighbour weights.

4.7 Layer 7: Bayesian Evidence Fusion Engine (BEFE)

BEFE pools three estimates on the logit scale with precision weights: the biological and transfer estimate (L1/L2), the tabular neural estimate (L3) and the graph estimate (L6). Base precisions are 1.0, 2.4 and 1.0, adjusted for availability, graph support and OOD status.

Uncertainty range (redesigned in 7.1). Two variances are added on the logit scale:

- the spread across simulated cycle scenarios, which falls to zero when the embryology is known;
- the variance of the pooled estimate under random-effects pooling (DerSimonian–Laird), using the precisions the fusion actually applied.

Disagreement between sources adds between-model heterogeneity τ^2 only where their own precision does not explain it. The range therefore narrows as observations arrive and widens when models genuinely disagree or the case is outside the reference domain. The displayed range is the pooled logit ± 1.96 combined SD, transformed to probability. It is a model-uncertainty range, not a confidence interval with validated coverage.

Reliability index. Consensus (weight 0.40), diffusion agreement (0.30), graph support (0.20) and phenotype certainty (0.10). An OOD alert caps the index.

OOD detection. Mahalanobis-distance checks are run separately on the clinical profile (age, AMH, AFC, BMI) and on the laboratory profile (oocytes, MII, 2PN, blastocysts, KPI). Coverage of these checks and the features they do not check are shown separately.

5. Additional clinical modules

5.1 Total Reproductive Potential (TRP): illustrative scenario projections

> All TRP outputs are labelled in the interface and in reports as **illustrative scenario projections**. They combine the current DT estimate with published relationships for AMH decline and cumulative success. They have not been validated as individual trajectories of future fertility or live birth, and they must not be presented as such.

TRP projects scenarios for a person's reproductive timeline. It computes:

- an individual AMH decline trajectory: log-linear with individual variability (Dölleman et al., Hum Reprod 2013); AFC is derived from AMH (Scheffer et al., Fertil Steril 2009);
- a Monte Carlo fan of possible future cycle sequences;
- importance resampling conditioned on the response observed in previous cycles;
- the cumulative probability of at least one clinical pregnancy over N cycles or N years;
- window-of-opportunity metrics (P10/P50/P90 of the age at which the per-cycle probability falls below a chosen threshold).

A per-attempt selection effect follows Malizia et al. (NEJM 2009; α = 0.08 per attempt), and the cumulative framing follows McLernon et al. (BMJ 2016). In the desktop application the starting probability is the displayed fused (L1–L7) per-cycle probability, including results already observed in the current cycle. Confirmed absence of a transfer in the current cycle does not set future attempts to zero.

The Android app uses a simplified form anchored to the scenario distribution of Layer 3 (KAN): P_cum(n) = 1 − (1 − P_KAN)^(n^0.7). The anchor is held constant across future cycles. This is an illustrative assumption, not an individual treatment-course forecast.

5.2 Embryo and oocyte banking

The banking section separates two quantities. The first is the target probability of pregnancy across transfers (reference points 50%, 70% and 80%). The second is the confidence of obtaining the required number of euploid embryos, computed with the Esteves approach. The outputs are the number of euploid embryos and the number of mature oocytes required for the chosen target and confidence. This section supports fertility-preservation counselling. It reports what would be needed, not what should be done. Like TRP, it is an illustrative scenario projection: the number of oocytes needed relies on literature-based stage rates and the local calibration of the DT.

5.3 Stimulation reference

The stimulation section summarises the expected ovarian response, a protocol type and a gonadotropin range from local guideline material. Follitropin delta has a separate first-cycle nomogram by AMH and body weight. This is reference information, not a prescription validated for the individual patient.

5.4 Clinic adaptation and laboratory compatibility

Centre calibration is an intercept correction on the logit scale; within-centre ranking does not change. Inclusion requires own oocytes, a single blastocyst transfer without PGT-A and a known clinical-pregnancy outcome. The data must contain at least 100 pregnancies, 100 negative outcomes and 100 patients. The interval of the correction accounts for repeated transfers of the same patient. A fingerprint of models, code and clinic statistics prevents reuse of a stale calibration. Neural networks are not retrained.

Laboratory compatibility needs at least 300 culture-observed cycles with 2PN ≥ 1. It reports OOD aggregates, observed-to-expected ratios of empty culture, and feature distributions. It does not change forecasts.

6. Laboratory quality and early warning

6.1 Three references that are never merged

Reference	Question	Basis
Relative	Has the laboratory changed relative to itself?	The laboratory's recent history (reference window)
Absolute	Where is the laboratory relative to the standard?	Vienna-consensus competency and benchmark levels
Model	Is the laboratory producing what these patients were expected to produce?	DT stage envelope frozen for each cycle before culture

A relative decline while the indicator stays within consensus limits is reported as a trend to check against case mix. A decline below the competency level prompts review of the laboratory protocol. Corroboration by an unusual culture window counts only if the window is unusual relative to the laboratory's own distribution of incubator excursions and media-lot changes.

6.2 Charts and thresholds

The system charts three indicators:

- fertilisation rate (2PN per inseminated MII);
- blastocyst development rate (blastocysts per 2PN);
- maturation rate (MII per oocyte).

Bernoulli CUSUM charts, tuned to detect a halving of the odds of success, run at culture closure (day 6). Supporting views are EWMA charts with overdispersion-inflated limits, monthly Shewhart p-charts with binomial and Laney p′ limits, and funnel plots. The oocyte-retrieval rate deliberately has no sequential chart, because its denominator (the follicle count) can drift with counting practice.

Thresholds are calibrated by parametric bootstrap (1,200 replicates) to a target number of false signals per year, not to an average run length. The targets are ≤ 24 watch and ≤ 2 alert signals per year for the three charts together. In the reference laboratory, the simulated system produced 23.97 false watch and 2.06 false alert signals per year. A halving of the odds of success was detected after a median of 0.9–1.4 months.

Incubator logs (MIRI multiroom, one row every 30 s) are converted into episodes: temperature, CO_2 and O_2 excursions, lid openings, alarms and logging gaps. Media lots are tracked in a laboratory database.

6.3 Case-mix-adjusted expected yield

For a period, stage or group of cycles, the DT gives the expected totals for the patients actually treated. In 59 rescored transfers of centre D, the DT expected 437.5 zygotes, 281.0 blastocysts and 183.0 good-quality blastocysts from oocyte and MII counts alone. The laboratory obtained 493, 268 and 219 (113%, 95% and 120%). The DT forecasts counts well but not individual stage fractions, so its role in process control is to provide expected totals. Individual control-limit adjustment by DT envelopes was tested, increased dispersion, and is disabled.

6.4 Language-model interpretation through MCP

Python computes every number and assigns every classification, such as KPI status or uncertainty band, before any language model is called. Agents receive a pseudonymised packet built from an explicit allow-list of analytic fields and may only explain what the packet contains.

- **Specialised roles** run on small models with narrow inputs and a ban on prescriptive vocabulary: KPI analyst, model auditor, culture-window and consumable forensics, per-cycle narrator.
- **A lead analyst** synthesises across packets.

- **A critic agent** receives the same packets and checks whether each statement follows from them. It looks for:
  - numbers that do not match the packet;
  - claims without basis;
  - hypotheses presented as conclusions;
  - configuration constants read as results;
  - proportions on degenerate denominators;
  - one measurement presented as several confirmations;
  - unmentioned confounders.

Unresolved high-severity findings block the report. The desktop application narrates locally with Gemma 4, and no cloud language model is used by default. The clinical faithfulness of narration has not yet been formally evaluated; a rating protocol for two specialists has been prepared.

7. Mobile application: DT × Laya 1.3.0

The Android app is a self-contained research application for one IVF cycle, intended for clinicians.

- **Inputs.** Cycle card or free-text description in English or Russian. The Laya encoder (a compact BERT-family classifier from convaiinnovations/laya, Apache 2.0, fine-tuned by the team) extracts 12 defined fields. High-confidence values confirmed by a number in the source text are entered into the card and remain editable. Uncertain values are computed by formulas, and conflicts with manual entries require clarification. Height and weight are entered, and BMI is computed.
- **Computation.** 2,000 reproducible DT scenarios (seed 42) for layers 1–3 hold known facts fixed and simulate the rest. Up to 64 weighted representative scenarios, selected by deterministic medoids in the 18-input space, are scored by Laya. The headline number is the weighted median.
- **Outputs.** Clinical-pregnancy probability with the central 95% of scenarios; histograms of "at least one pregnancy per cycle" and "pregnancy with one embryo transfer"; the shares of scenarios without a transfer, without blastocysts and without good blastocysts; cycle pathway; TRP outlook.
- **Privacy.** No INTERNET permission. History is saved only on request, encrypted with AES-GCM and a key in the Android Keystore. System backup is disabled. JSON export is manual only.
- **Verification.** 12 states match the source demo; KAN formula, TRP, zero handling, fixed facts, reproducibility, tokenizer and serialisation are tested. The PyTorch to ONNX conversion was checked. A native run on Android 10 x86_64 confirmed model loading, the scenario pipeline, Russian extraction, encrypted history and the absence of INTERNET permission.
- **Scope.** FT-Transformer, CSDI, GAT, OOD, Layers 4–7 and BEFE are not executed on the phone. Laya on DT scenarios is a dependent part of the pipeline, not an independent confirmation of the DT. Testing on physical ARM64 phones (RAM use, speed) is a separate acceptance step.



Figure 2. Product screens in demonstration mode with synthetic data. Left: DT 7.1 clinician interface before treatment. Right: Android app, on-device scenario histograms.

8. Addressing the Reproductive Potential challenge

8.1 Mapping to the six requirements

Requirement	How the DT addresses it	Status
1. Longitudinal view of reproductive potential, direction of change	Stage-by-stage updating within a cycle; immutable revision history; illustrative TRP scenario projections over years, labelled as such; uses only the inputs provided, with missing and imputed values explicit	Available
2. Plain-language interpretation: modifiable versus non-modifiable factors, evidence, uncertainty	Each input tagged measured, missing or imputed; fixed factors (age, ovarian reserve) shown separately from modifiable ones (BMI, timing of a decision, number of cycles); uncertainty range, reliability index, OOD flags; local LLM explains pre-computed numbers only	Available; person-facing wording in the pilot
3. Timely, personalised options within intended use	Banking targets for 50, 70 or 80% chance; illustrative multi-cycle scenarios; stimulation reference; PGT-A only as a separate embryo-selection scenario with no implied gain in cumulative live birth; no prescribing	Available
4. Clinician and system usefulness	One-page clinician report; case-mix-adjusted laboratory KPIs; early warning; aggregated expected-versus-observed by age band and centre via API and MCP	Laboratory available; population dashboard in the pilot
5. Privacy, non-discrimination, user control	On-premise and on-device processing; keyed-hash pseudonyms; allow-listed agent fields; licence prohibits employment, insurance and eligibility use; granular, revocable consent added in the pilot	Core available; consent controls in the pilot
6. Evidence of earlier action, calibration and false-reassurance risk	Calibration and false-reassurance measured and corrected (section 9.5); pilot measures time from risk signal to counselling, decision quality, referral completion and equitable reach	Pilot

8.2 Main clinical scenario: the first consultation

The demonstration and the pilot centre on one scenario.

Step	What happens	Implemented in 7.1
1. Patient arrives	Age, AMH, AFC and history are available; nothing else is required	New calculation form; missing fields stay empty and are flagged
2. Twin computes ranges	Distributions of oocytes, MII, blastocysts and possible euploid embryos, with an uncertainty range, a reliability index and missing-data flags	Results: clinical and extended views
3. Clinician compares scenarios	Treatment now; delayed treatment (recalculated at a later age); several cycles (TRP); oocyte cryopreservation (banking). All are labelled illustrative scenario projections	Research view: TRP and banking sections; recalculation with a changed age
4. Assumptions shown	Which inputs were measured or imputed, which literature relationships the scenario uses, and what should be checked next (for example a measured AMH instead of a default)	Calculation details, input provenance, OOD flags
5. Shared decision	Plain-language explanation of the computed results for the patient; the clinician decides with her	Local LLM interpretation window; PDF report (EN/RU)

Today these steps use separate sections of the 7.1 application. The pilot deliverable is one **first-consultation panel** that places the scenarios side by side, lists their assumptions and produces the patient summary. The panel reuses the 7.1 computation path and adds no new model. A consent screen lets the person choose which inputs are used and withdraw them later; withdrawn inputs are removed from the stored revision and the forecast is recomputed.

8.3 Alignment with Abu Dhabi priorities

The Abu Dhabi Programme for the Future of Reproductive Health calls for awareness and prevention across life stages, connected care, and technology introduced with evidence and governance. The DT contributes in three ways:

- an evidence-led, calibrated view of reproductive potential that can be offered earlier than a treatment decision;
- case-mix-adjusted expected values that complement the age-stratified outcome reporting required of licensed IVF providers, so centres treating different patients can be compared fairly;
- an auditable, on-premise architecture that is consistent with the DoH Policy on Use of AI in Healthcare, its Responsible AI Standard, and federal health-data law.

9. Validation and performance

9.1 Evidence streams

Stream	Population	Design	Release
Development and held-out test	16,056 cycles, 2013–2023, six countries; held-out 1,606	Fixed random split	Components of 7.0/7.1
Calibration set	1,556 cycles, 2024–2025, single centre	Calibration, fusion weights, phenotypes	7.0/7.1
Prospective clinical cohort	Centre D (Tashkent), 59 transfers (26 pregnancies), Feb–Sep 2026	Forecasts generated during treatment and disclosed only after transfer; rescored with locked 7.1 through the API, no adaptation	7.1 (fingerprint 91704e29)
Temporal validation	Centre E (Almaty), new data source onboarded after the DT was locked; 1,604 fresh transfers 2023–2025 and 234 in 2026	Year-by-year evaluation; 2026 kept separate	7.1
Extended clinical variables	Centre E, 2023–2025 development, 2026 test	Protocol frozen before the 2026 cohort was scored	7.1 anchor
Image experiments	2,344 public images for development; 300 day-5 blastocysts at centre B for test	Same structured-data core with or without image features	Research
Culture risk and euploidy	1,050 culture-observed cycles (centre B)	Retrospective	7.1

9.2 Pregnancy prediction

Evaluation	AUC (95% CI)	Brier
Prospective clinical cohort, headline (transfer view)	0.761 (0.629–0.878)	0.213
Prospective clinical cohort, retrieval-only view	0.754 (0.625–0.873)	–
Centre E, 2026 temporal validation, all fresh transfers	0.702 (0.622–0.771)	–
Centre E, 2023–2025, all fresh transfers	0.640 (0.612–0.668); 0.637–0.639 in each year	–
Held-out development test, KAT	0.820 (0.798–0.840)	0.163

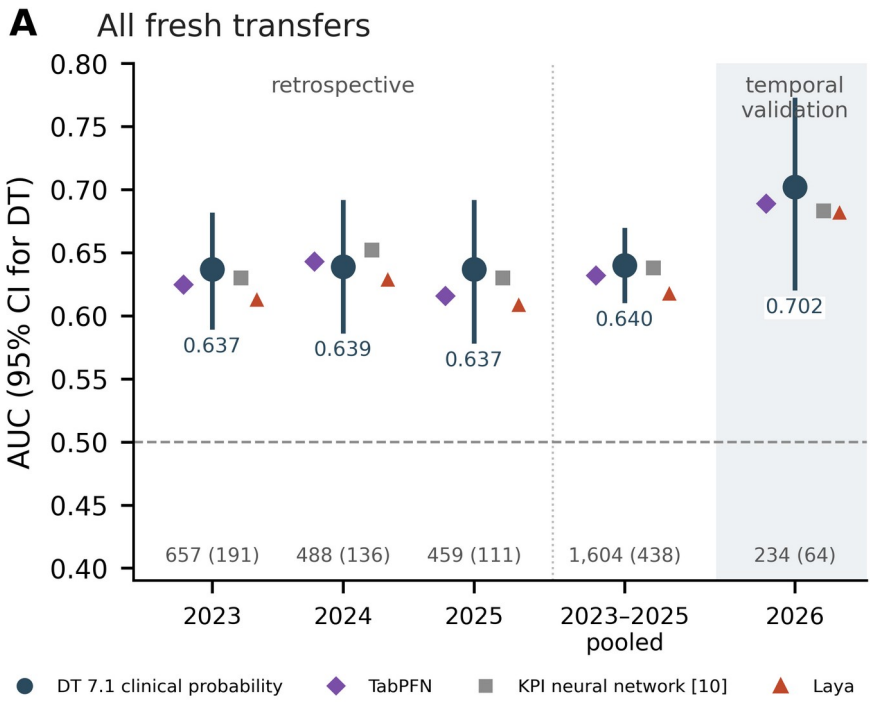


Figure 3. Centre E: AUC of the DT 7.1 clinical probability (circles with 95% bootstrap CI) and of TabPFN, the KPI neural network and fine-tuned Laya, by year, for all fresh transfers. The 2026 temporal-validation cohort (shaded) was scored with the same locked release and is not pooled with earlier years.

At centre E, pre-trained comparators reached similar values in every year and subset; paired differences from the DT ranged from −0.035 to +0.016.

Comparison with simple baseline models. The DT was locked before centre E was onboarded and was not trained on its data. The comparison models in the second row were trained once on 1,610 fresh transfers of the same centre (2023–2025) and evaluated once on the 2026 cohort, using the DT laboratory features with AMH and BMI.

Model (2026 test, centre E)	AUC
DT 7.1, locked, no local training	0.702
Trained on the centre's own data: logistic regression (L2, C = 0.1) · gradient boosting · TabPFN	0.732 · 0.656 · 0.743
Pre-trained, no local training: TabPFN · KPI neural network · Laya (fine-tuned elsewhere)	0.689 · 0.683 · 0.682

Without local training, the locked DT therefore performs in the range of simple models fitted to the clinic's own history. In addition, it provides stage-level forecasts and uncertainty, and it can be used before a clinic has accumulated a training set. When a logistic regression fitted at one centre was moved to another (six directed transfers between centres A–C), its worst destination Brier score was 0.547, against 0.264 for the unchanged DT. The regression could be applied to only 22.6% of cycles (those with measured AMH), whereas the DT returned an estimate for every cycle, with substitutions flagged.

9.3 Stage forecasting

In the prospective clinical cohort, a DT view that knew only oocyte and MII counts forecast the stage counts as follows. The median absolute error was one embryo at each stage.

Stage	Spearman ρ (95% CI)	Inside 95% interval
2PN zygotes	0.961 (0.927–0.977)	88%
Blastocysts	0.704 (0.521–0.837)	97%
Good-quality blastocysts	0.732 (0.572–0.842)	98%

Across 5,205 historical stage observations from four centres, usable P10–P90 envelopes contained 81.2% of observations, close to the nominal 80%.

9.4 What does not add information: images and extended clinical data

Embryo images. Image features were added to the same structured-data core and tested on 300 day-5 blastocysts.

Model	Test AUC
Structured data, KAT (2,000-transfer set containing the panel)	0.75
+ ResNet34 features	0.67
+ DINO ViT features	0.78
+ masked-autoencoder ViT-B/16	0.61
Inner cell mass and trophectoderm images only	0.74

The encoder ranking reversed between validation and test. Attribution maps of the two architectures agreed poorly (SSIM 0.388), and transformer maps changed after small perturbations. An image branch would therefore reduce the transparency on which the DT depends.

Time-lapse versus benchtop culture. Clinical pregnancy after blastocyst transfer did not differ between time-lapse, MIRI benchtop and Cook benchtop incubation (χ^2 p = 0.80; adjusted OR for time-lapse 1.03, 95% CI 0.79–1.35). This is consistent with randomised trials (TILT, Lancet 2024; SelecTIMO, Lancet 2023) and with the randomised evaluation of deep-learning embryo selection (Illingworth et al., Nat Med 2024).

114 extended clinical variables. These were grouped a priori into four blocks: hormones (12), stimulation (23), transfer and endometrium (22), and history (57).

- In a DT-anchored logistic model, all blocks together added no information to the DT (likelihood-ratio p = 0.80).
- In the 2026 test, no block improved the DT (AUC 0.702).
- For TabPFN, all blocks together changed the AUC from 0.744 to 0.773 (ΔAUC +0.029, 95% CI −0.001 to 0.063), which was not significant.
- The only consistent information concerned the transferred embryo (embryo stage at transfer and PGT status). Both variables are now used in the analytics pipeline after the DT prediction.

Taken together: routinely recorded tabular data are sufficient for the forecast, and no imaging hardware or extended data capture is needed.

9.5 Calibration and false-reassurance risk

In the prospective cohort, without local recalibration, the mean predicted per-transfer probability was 53.0% against 44.1% observed (+8.9 percentage points). By quintile of prediction, the mean prediction versus the observed rate was 0.33 vs 0.17, 0.47 vs 0.17, 0.53 vs 0.55, 0.62 vs 0.58 and 0.71 vs 0.75. Over-prediction was therefore concentrated in the lowest two quintiles. At centre E, calibration slopes of the different outputs ranged from about 0.1 to 0.8 between years, which is why local calibration review is a gate before any clinician-facing use.

Finding	Correction
All models over-predicted pregnancy at centre E (DT O/E 0.73–0.81 among transfers with a blastocyst)	A centre intercept fitted on other years restored O/E to 0.99–1.02 in each held-out year
Risk of no blastocyst: good ranking (AUC 0.824, 95% CI 0.774–0.875) but under-predicted (1.6% predicted vs 9.3% observed), a false-reassurance risk	Cross-fitted local recalibration reduced the Brier score from 0.079 to 0.066. The pilot reports observed vs predicted adverse outcomes in low-risk groups every quarter
Euploid share under-estimated (61.2% predicted vs 78.1% observed); discrimination AUC 0.712	Age-stratified prior to be re-fitted with local PGT-A data

Finding	Correction
Sample needed for a reliable intercept depends on the pregnancy rate	About 70, 226 and 458 transfers for 80% coverage within 10% at pregnancy rates of 70%, 42% and 26%; the application requires ≥ 100 pregnancies and ≥ 100 negatives

9.6 Laboratory monitoring

In the reference laboratory (102 closed cycles, December 2025 to September 2026), the CUSUM charts raised 17 watch episodes and no alert. EWMA charts stayed within overdispersion-inflated limits, and monthly p-charts had no point outside their limits. Simulated detection delays were:

- halving of the odds of success: 0.9–1.4 months;
- odds reduced to 0.7 of baseline: 4.3–5.9 months;
- odds reduced to 0.8 of baseline: 20–32 months.

Slow declines are therefore detected late, a property the system reports rather than hides.

10. Software verification and quality assurance

Check	Result
Numerical equivalence of source and protected build	7 scenarios, 831,816 numeric values, maximum absolute deviation 0
Regression tests	341 passed; two remarks fixed and re-verified with 67 targeted tests
MCP connector	67 tests including 19 security tests; a real Codex CLI client connected and received all 11 tools
Installer	Installed into a folder with a Cyrillic path; manifest signature, 15,392 payload files, library imports and 17 encrypted models verified
Integration service	HTTPS, certificate verification, scopes, key revocation, DPAPI under both accounts; analytics result equal to direct computation
Reproducibility of stored runs	Four 7.1 reruns returned identical outputs for all jointly scored cycles (36 of 36); five deterministic artefact types byte-identical
Synthetic snapshot contract tests	21 of 21 expected responses
Data-orchestration defects found during development	17 silent defects that altered data, denominators or cohorts without stopping a run; all corrected and covered by contracts
Android	Formula, scenario, convergence, tokenizer, ONNX conversion and native emulator checks (section 7)

The checks were run on the developer's Windows PC with a separate installation; a clean Windows virtual machine was not available. These checks are not a formal medical-software certification or a penetration test.

11. Security, privacy and data governance

11.1 Data protection by design

- **Local processing.** Desktop computation runs locally. The desktop HTTP server binds to 127.0.0.1 with a per-launch token and Host-header checks. The integration service opens no outbound connections, accepts no callback URLs and uses no external AI services.
- **Pseudonymisation.** Cycle identifiers are keyed hashes (HMAC) of the clinic code with a local secret. The crosswalk and API keys are protected with Windows DPAPI. Names, clinic codes, dates, staff names and free text never appear in MCP responses, and every response is checked before it is sent.
- **Data minimisation.** Agents read only an allow-listed set of numeric analytic fields. Identifiers are re-attached locally by Python after interpretation. Uploaded tables are processed in memory and are not written to the application directory.
- **Encryption.** History is protected by user-bound Windows protection on desktop and by AES-GCM with an Android Keystore key on mobile. Integration snapshots use machine DPAPI with restrictive ACLs. Models are encrypted (AES-GCM), and the manifest is signed (Ed25519, SHA-256 hashes).
- **Cloud agents.** If a clinic connects a cloud AI agent through MCP, numeric features and forecasts reach that agent's provider. This is documented, MCP is not enabled automatically, and the default narration is local.

11.2 Network security of the integration service

The service enforces the following controls:

- TLS 1.2 or later, with certificate verification.
- Loopback or one explicitly selected private IPv4 interface; up to 25 allow-listed client IPs, checked at the TCP boundary.
- Windows Firewall scoping; no public or wildcard addresses.
- One-time bearer secrets, stored only as SHA-256 hashes, with rotation and revocation.
- Bounded request sizes, concurrency limit 12, and 120 requests per minute per IP.
- Separate `mis` and `analytics` scopes. Clinic-system results are computed only after clinician review. Analytics results are marked as not clinician-reviewed and are not clinical forecasts.

11.3 Governance in the pilot

- Data stay in the pilot clinic.
- Use follows DoH research approval and the clinic's consent process.
- Aggregated outputs for the health system contain no individual records.
- Quarterly audits of calibration, false reassurance, OOD frequency and subgroup performance are reported to the clinic's AI governance lead.

12. Responsible AI

Principle	Implementation
Human oversight	A qualified clinician reviews every output before it informs care or reaches a patient. The system takes no autonomous actions
Transparency	Stage-by-stage envelopes; fusion breakdown showing how much of the answer came from biological mechanism and how much from data; reliability index; coefficients traceable to cited sources
Honest uncertainty	Model-uncertainty range labelled as not a validated confidence interval; imputed inputs flagged; degenerate envelopes detected; abstention distinguished from error
Fairness and transport	Local calibration gate; OOD flags; calibration and AUC reported per age band and centre; between-centre variability treated as a finding to report
Language-model safety	Code computes and classifies; models explain allow-listed packets; prescriptive vocabulary banned; critic agent blocks unsupported claims
Non-discrimination	Probabilities with ranges instead of labels; licence and pilot terms prohibit use in employment, insurance or eligibility decisions
Accountability	Immutable revisions, run manifests, model fingerprints and logs support audit

13. Assumptions and limitations

13.1 Clinical scope

- The endpoint is clinical pregnancy with fetal cardiac activity, not live birth.
- Forecasting from the first consultation is a design capability. Pregnancy evaluations reported here begin at oocyte retrieval or later, and the pre-treatment pathway will first be evaluated in the pilot.
- No evidence yet shows earlier counselling, better decisions or improved outcomes. These are pilot endpoints. Whether AI improves clinical outcomes requires prospective, ideally randomised, evaluation.
- The prospective clinical cohort is small (59 transfers, wide confidence interval) and comes from one centre with younger patients. AMH was imputed in 53 of 59 cycles.

13.2 Model assumptions

- Layer 1 imports per-oocyte maturation and fertilisation coefficients from a published model. The euploidy branch uses a literature prior that under-estimated the euploid share in local data. Warming survival is fixed at 0.95.
- The Layer 2 clinical coefficients were specified for the platform, and only its intercept was recalibrated locally.
- BEFE treats the L1/L2, KAT and GAT estimates as separate sources. They are strongly correlated (pairwise logit correlations 0.64–0.80; about 1.42 effective independent estimates out of three). Fusion improves interpretability more than discrimination.
- The provenance of the exact fit of the deployed isotonic calibrator is not fully reconstructed. Control outputs before and after the 7.1 correction matched (Brier 0.1594, ECE 0.0279 on 1,606 held-out rows).
- TRP and banking outputs are illustrative scenario projections. They rely on literature AMH-decline and cumulative-success relationships, and the mobile TRP keeps the current anchor constant over future cycles. They are not validated individual trajectories and must not be presented as forecasts of future fertility or live birth.
- PGT-A is represented only as a scenario of embryo selection and use. The model does not imply that PGT-A increases cumulative live birth.

13.3 Transport between centres

- Discrimination and calibration differ between centres and data paths. Absolute probabilities can be biased until a local intercept is fitted. The intercept needs roughly 70–460 transfers depending on the pregnancy rate, and it does not change within-centre ranking.
- Routine predictors carry limited patient-level information about pregnancy. Model rankings in development data did not hold externally, so an architecture should not be chosen for its development AUC alone.
- The representation of identical records can change the performance of comparator foundation models. Mapping versions are therefore recorded together with model versions.

13.4 Data

- Complete inputs do not imply complete measurements. Defaults and imputations are flagged, and forecasts based on them describe a partly hypothetical patient.
- Stage forecasts are reliable for counts but weak for individual stage fractions. The DT is therefore used for expected totals in quality control, not for adjusting single-cycle control limits.
- Process-control frequencies and detection delays are simulation estimates under the stated baseline, volume and dispersion. Slow declines are detected late.

13.5 Software and deployment

- The software is not a medical device. Routine clinical use in the UAE would require registration with the Emirates Drug Establishment as decision-support software, and DoH approvals.
- The Windows installer has no Authenticode publisher signature yet. Clean-machine installer acceptance and a formal penetration test are pending.
- The interface is in English and Russian; Arabic is planned.
- The Android app runs a reduced pipeline (Layers 1–3 plus Laya). Physical-device acceptance (RAM, speed) is pending, and the app is side-loaded, not distributed through an app store.
- The clinical faithfulness of language-model explanations has not been evaluated. Generated text can be wrong and must be reviewed by a clinician.

14. Changes from version 7.0 to 7.1

Area	v7.0	v7.1
Retrieved oocytes (L1)	Zero-inflated negative binomial	Negative binomial with log link, fitted by zero-truncated likelihood with five-fold patient-level cross-validation; coefficients updated 11 Sep 2026
Blastocyst stages (L1)	Binomial filters with age-dependent literature rates	Zero-inflated beta-binomial (blastocysts) and beta-binomial (good quality) with dispersion; fitted on 7,919 culture-observed cycles
Several transfers per retrieval	Not described	Shared logit-normal cycle effect σ = 0.746; "transfers planned" limits the per-cycle estimate
Fair vs good blastocyst	Not described	Adjusted OR 0.51 (0.35–0.75)
KAT calibration	Venn–Abers wrapper	Isotonic calibration with interpolation between step centres (older Venn–Abers wrappers still load)
FT-Transformer encoding	PLE depended on the batch maximum	Trained bin boundaries preserved; single and batch scoring identical (deviation < 1.1×10^{-7} after the fix, previously up to 0.120)
Uncertainty range	Scenario spread plus total weighted source spread; widened with each new observation	Scenario variance plus random-effects pooled variance; narrows with observations, widens with genuine disagreement or OOD
CSDI handling	Equifinality check with Layer 1	Intentional abstention separated from errors; erroneous or OOD outputs excluded from fusion; role defined as a consistency check
Clinic adaptation	Beta-binomial clinic posterior (v7.0, section 7)	Intercept recalibration with inclusion criteria and minimum sample (≥ 100 pregnancies, ≥ 100 negatives, ≥ 100 patients); fingerprint protection
Laboratory compatibility	–	Report from ≥ 300 culture-observed cycles: OOD, empty-culture O/E, feature distributions
Application	Streamlit research app	Native desktop app (WebView2), EN/RU interface and PDF, encrypted history, CSV/XLSX batch input (5 MB; 200 rows batch, 20,000 rows analysis)
Integration	–	Windows service with HTTPS REST API v1, OpenAPI specification, clinician inbox, analytics scope
AI agents	–	MCP connector with 11 tools, privacy-checked responses, plugin for Claude Code and Codex
Local language model	Not part of the v7.0 documentation	Gemma 4 through the bundled Ollama engine, separate interpretation window and print
Packaging	Source distribution	Offline installer with bundled runtime, 17 encrypted models, signed manifest, protected compiled modules
Mobile	–	DT × Laya Android 1.3.0, offline
Laboratory quality	Control-charting concept (v7.0, section 16)	Early Warning System with three references, calibrated thresholds, incubator and lot forensics

15. Roadmap and pilot plan

Phase	Period	Activities	Exit criteria
0. Readiness	Dec 2026–Jan 2027	Partner IVF centre in Abu Dhabi; research approval; data audit with orchestration contracts; on-premise installation	Data contracts pass; approvals in place
1. Local calibration	Months 1–3	Retrospective run; centre intercept; calibration and false-reassurance report by age band; first-consultation panel completed	O/E 0.9–1.1 per age band; false-reassurance report accepted
2. Silent prospective	Months 3–6	Forecasts stored, not shown; stage coverage; early-warning replay; usability testing	≥ 80% of stage counts inside P10–P90; SUS ≥ 70
3. Supervised use	Months 6–12	Clinician-facing counselling including pre-treatment consults; measure time to counselling, decision quality, referral completion, equity	Pre-specified pathway targets
4. Scale	Months 12–24	Additional centres; Arabic interface; FHIR mapping and Malaffi-linked inputs; aggregated DoH dashboard; EDE registration pathway	Multi-centre calibration; regulatory dossier

15.1 What the pilot clinic receives

- First-consultation counselling: ranges, illustrative scenarios and a plain-language summary.
- Per-cycle stage forecasts with deviation flags for the embryologist.
- A monthly laboratory report: Vienna-consensus KPIs, expected-versus-observed yield, early-warning charts.
- A local calibration report (O/E by age band) and a false-reassurance audit before any clinician-facing use.
- A data-quality audit of its own exports (silent defects, denominators, defaults).
- Governed AI summaries through MCP, with pseudonymised data.
- An independent evaluation report at the end of the pilot.

15.2 Resources the pilot needs (planning estimates)

Resource	Estimate
Retrospective data	2–3 years of cycle exports; at least 100 pregnancies and 100 negative outcomes among single-blastocyst transfers (own oocytes, no PGT-A) for the centre intercept; at least 300 culture-observed cycles for the laboratory compatibility report
Prospective data	About 200 first consultations and 250 transfers over 9–12 months
Integration	Offline installation: 1 day per workstation. Data mapping and audit: 2–4 weeks. Optional EMR/LIS connection through REST API v1: 4–6 weeks
Embryologist	2–4 hours per week: data review, deviation flags, early-warning signals
Clinician lead	About 2 hours per week: counselling workflow, decision-quality measures
IT specialist	3–5 days at set-up, then about 2 hours per month
Hardware	Existing Windows workstation, CPU only; no new devices

15.3 From working prototype to independent clinical pilot

The pilot moves from the current working prototype to an independent evaluation in five steps:

1. The model version is locked and its fingerprint recorded.
2. The protocol, endpoints and analysis plan are registered before prospective data collection.
3. A silent prospective phase precedes any exposure of forecasts to clinicians or patients.
4. Analysis is performed by an independent evaluator, for example MBZUAI or the clinic's statistician.

5. The DT is compared with a locally fitted simple model and with usual counselling.

The figures in section 15.2 are planning estimates to be agreed with the pilot clinic.

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